

< T E S T >

3 hours, Closed Book with Calculator Allowed

Nov. 29, 2017

Problem 1 [20 pt]

Barges arrive at a lock at an average of 4 each hour.

- (1) If the arrival of barges at the lock can be considered to follow a Poisson process, what is the probability that 6 barges will arrive in 2 hours?
- (2) If the lock master has just locked through all of the barges at the lock, what is the probability he can take a 15-minute break without another barge arriving?
- (3) If the operation of the lock is such that 4 barges can be locked through at once and the lock master insists that this always be the case, what is the probability that the first barge to arrive after 4 previous barges have been lock through will have to wait at least 1 hour before being locked through?
- (4) Derive the Poisson pmf from the binomial distribution.

Problem 2 [30 pt]

A river with annual flows $X \sim N(240, 40)$ is joined by a major tributary with annual flows $Y \sim N(150, 50)$ at point P. At point Q on the river below P there is a diversion with annual flows $Z \sim N(100, 25)$. Below Q, suppose the annual flows are denoted by R. If X, Y, and Z are independent,

- (1) Determine the distribution of R.
- (2) Calculate the 100-year flood below Q.
- (3) & (4) Recalculate (1) and (2) and if X and Y are correlated with their correlation coefficient be 0.5. Make your comment on the effect of the correlation on the above results.
- (5) Determine the conditional pdf of X when $Y = 100$ and 200 when X and Y are still correlated with their correlation coefficient be 0.5. Make your comment.

Problem 3 [15 pt]

- (1) In order to be 90% percent sure that a design storm is not exceeded in a 10-year period, what should be the return period of the design storm?
- (2) Estimate the chance of at least one occurrence of a T-year event in T years. Make your comment.

Problem 4 [20pt]

Hydraulic engineers need to estimate the annual maximum flood flow exceeded with a 0.1% probability (a chance of 1 in 1000) for a reservoir's spillway. If such floods have a Gumbel distribution with mean 1500 cms and standard deviation 500 cms,

- (1) what is the design value they seek?
- (2) Why is a Gumbel distribution a reasonable choice for describing floods?
- (3) Calculate the return period T of the mean annual flood under the Gumbel distribution?

Problem 5 [15pt]

- (1) Derive the Yule-Walker equation of the AR(p) model.
- (2) Write the 3-month ahead forecast using the ARMA(1,2) model. The notations you use should be clearly defined.

Good Luck!